# Day 19 Of Communication System Class

**9 Noise in Communication Systems**
**9.1 Definition, white noise, AWGN channel, PSDF, and AC function of white noise**
**9.2 Ideal low-pass and RC filtering of white noise, noise equivalent bandwidth of a filter**
**9.3 Optimum detection of a pulse in additive white noise, the matched filter, realization of matched filters (Time correlators), the matched filter for a rectangular pulse**
**9.4 Overview of error probability function in digital communication (ASK, FSK, and PSK)**

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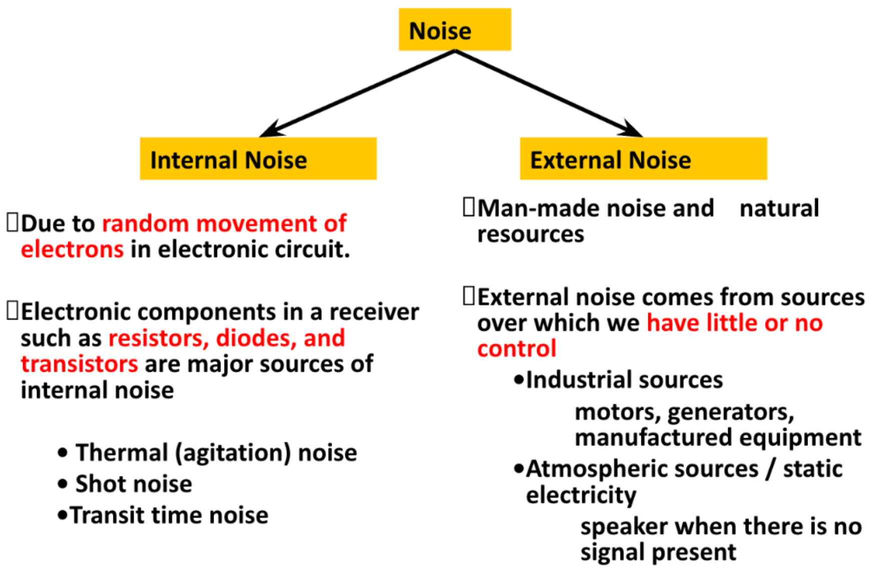
## 9.1 Definition, white noise, AWGN channel, PSDF, and AC function of white noise

Noise is a general term which is used to describe an "unwanted signal which affects a wanted signal."

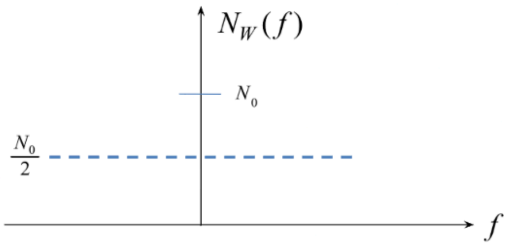
**Noise is a random signal that exists in a communication system.**

**Random signal cannot be represented with a simple equation.**

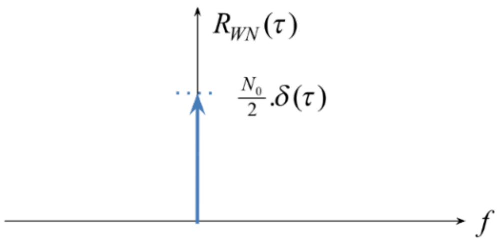


White noise is an ideal case of description of noise in communication systems. The power spectral density of white noise is independent of frequency. i.e. it has flat spectral density, $N_W(f)$ over $-\infty < f < \infty$ and has zero mean value.

$$PSDF\ of\ white\ noise = \begin{cases} \frac{N_0}{2} & for -\infty < f < \infty \\ N_0 & for\ 0 < f < \infty \end{cases}$$




AC function of white noise is therefore a delta function.



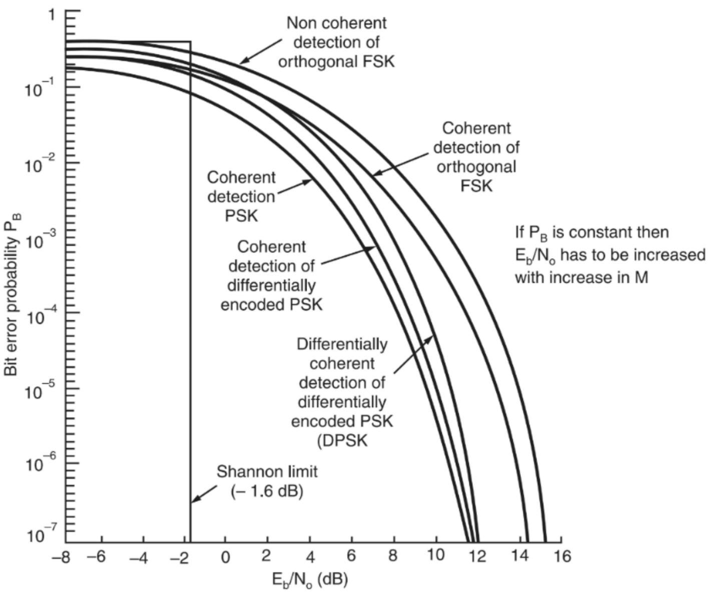

As $R_{WN}(\tau) = 0$; for $\tau \neq 0$
Two different samples of white noise no matter how close they are in time shift($\tau \rightarrow 0$) are uncorrelated.
The origin of term "white noise" is white light where the frequencies of all the color exists. It has no physical significance as it doesn't exist in nature but has convenient mathematical properties so useful for system analysis.

**9.2 Ideal low-pass and RC filtering of white noise, noise equivalent bandwidth of a filter**

**9.3 Optimum detection of a pulse in additive white noise, the matched filter, realization of matched filters (Time correlators), the matched filter for a rectangular pulse**

## 9.4 PERFORMANCE COMPARISON OF DIGITAL MODULATION SCHEMES USING A SINGLE CARRIER

| S. No. | Name of the system | Error probability |
|---|---|---|
| 1. | Coherent BPSK | $(1/2)\ \text{erfc}\ \sqrt{E_b/N_0}$ |
| 2. | Coherent BFSK | $(1/2)\ \text{erfc}\ \left(\sqrt{E_b/2N_0}\right)$ |
| 3. | DPSK | $(1/2)\ e^{-(E_b/2N_0)}$ |
| 4. | Non-coherent FSK | $\left(\frac{1}{2}\right)e^{-(E_b/N_0)}$ |
| 5. | QPSK | $\text{erfc}\left(\sqrt{E_b/N_0}\right)-\frac{1}{4}\,\text{efrc}^2\left(\sqrt{E_b/N_0}\right)$ |
| 6. | MSK | |



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